

Fatma Nur Karagöz

Ankara, Türkiye | fk6895164@gmail.com | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Work-Based Training / Internship Availability: February 8 – May 30 (Spring Semester of 4th Year)

PROFESSIONAL SUMMARY

I have just moved up to my 4th year of Software Engineering and am an aspiring Full-Stack Developer who builds user-friendly, high-performance and scalable applications with modern web technologies. While deepening my expertise in the React, TypeScript and Node.js ecosystem, I embrace clean code, algorithm analysis, responsive interface development and strong user experience (UI/UX) design principles. Alongside software development, I take an active interest in scientific research in Large Language Models (LLMs) and Data Science, turning AI solutions into practical applications through academic projects. With my analytical mindset and strong team coordination, I aim to grow my professional experience as an intern/engineer candidate who can develop both myself and my team.

EXPERIENCE

Software Developer Intern – ÇAKÜ Department of Information Technology **July 2026 – September 2026**

During my summer internship, I designed and developed StudyMentor (an AI-Powered Learning Analytics Platform) end to end; I was responsible for the entire frontend, backend, ML microservice and deployment process.

Volunteer Intern – Republic of Türkiye Ministry of Health – Ankara

August 2024

Worked with a mentor on frontend and backend technologies (JavaScript libraries). Gained hands-on experience by observing user interface development, API integration and system design processes in web applications.

PROJECTS

StudyMentor — AI-Powered Learning Analytics Platform

Status: Live

I developed an AI mentor application that analyzes students' study routines, focus durations and moods to provide personalized study and productivity guidance.

- Developed an AI microservice with Python (FastAPI, Scikit-Learn, Pandas) that analyzes users' study data.
- Designed and developed the responsive interface with React, TypeScript and Tailwind CSS.
- Built a modular and scalable backend architecture using Node.js (Express), PostgreSQL (Supabase) and Prisma ORM.
- Containerized the backend API with Docker and deployed it automatically to Azure Container Apps through a CI/CD pipeline with OIDC-based (passwordless) authentication; published the frontend on Vercel.
- Technologies Used: React, TypeScript, Tailwind CSS, Node.js, Express, PostgreSQL, Supabase, Prisma ORM, Python, FastAPI, Scikit-Learn, Pandas, Docker, Azure, GitHub Actions (CI/CD, OIDC), Vercel
- [StudyMentor Website Link](#)

TART — Critique Community

Status: Live

I designed and developed the full-stack architecture of a web and mobile (cross-platform) critique community platform for young people.

- Established a layered API architecture with Node.js, Express, Prisma ORM and PostgreSQL, securing the database with Supabase RLS.
- Developed the React (Web) and Expo / React Native (Mobile) interfaces in TypeScript, synchronized with a single backend service.
- Developed comment voting, nested replies, a follow system and real-time in-app notification mechanisms for user interactions.
- Integrated Azure CI/CD processes, automating the testing and deployment stages.
- Technologies Used: React, React Native (Expo), TypeScript, Node.js, Express, Prisma ORM, PostgreSQL, Azure Pipelines, Framer Motion
- [TART Website Link](#)

TE4IT — Task & Education for IT

Status: MVP Completed

I took an active role in the frontend development of an enterprise platform that combines project management with educational content for software teams and IT students.

- Developed a modular and scalable component structure with React.
- Designed role-based screen management and dynamic interface flows.
- Implemented RESTful API integrations.
- Made UI/UX improvements and user experience optimizations.
- Worked in coordination with the backend team to build the interface architecture of the task management module.
- Technologies Used: React, JavaScript, Axios, Context API, REST API
- [TE4IT Website Link](#)

LLM-Based Supercapacitor Literature Analysis System

AI and Data Science Researcher (TÜBİTAK STAR Scholar Program)

- I am continuing work on designing an analysis system that uses Large Language Models (LLMs) and RAG (Retrieval-Augmented Generation) architectures to automate the analysis of supercapacitor literature in the field of energy storage.
- I manage data mining, semantic analysis and the transformation of unstructured data from academic texts into datasets.
- Technologies Used: Python, LLMs, RAG, Data Science & Text Mining Tools, Google Colab.

Scan & Serve – Smart QR Menu and Ordering System

I developed a QR code-based digital menu and ordering system for restaurants.

- Designed and developed the user interface with React.
- Structured the order flows and state management.
- Integrated with a Spring Boot backend.
- Established connections to the MongoDB database.

Hospital Automation System

I developed a Windows Forms-based automation system using C#, SQL Server and DevExpress.

- Designed the patient registration, appointment and reporting modules and applied a layered architecture approach.

Unmanned Underwater Vehicle – Ukulan (Teknofest)

I took part in the software development of an autonomous underwater vehicle.

- Developed a simulation environment with ROS and Gazebo and worked on autonomous control algorithms.

EDUCATION

Firat University – Faculty of Technology – Software Engineering (2023 – Present)

GPA: 3.27 / 4.00

SKILLS

Frontend: React, JavaScript, TypeScript, HTML, CSS, Responsive Design

Backend: Node.js, Spring Boot, C#, Java

Data Science & AI: Large Language Models (LLMs), RAG Architecture, Python, Data Analysis, Google Colab, Jupyter Notebook

Databases & Cloud: SQL Server, MongoDB, PostgreSQL, Supabase

Tools & Methodologies: Git, GitHub, REST API, UI/UX (Figma), UML, Agile/Scrum, CI/CD (Azure Pipelines), Docker, Deployment Processes

Engineering Approach: Commitment to clean code and sustainable software principles, adaptation to an Agile working culture, systematic thinking and a modular development approach.